

Aletheia: An Offline-First Clinical Decision Support System for Differential Diagnosis in Low-Resource Healthcare Settings

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Abstract—Access to specialist clinical expertise remains severely limited across sub-Saharan Africa, where physician-to-patient ratios can fall below 1:25,000 in rural settings. Existing AI-assisted diagnostic tools predominantly require reliable internet connectivity and high-specification hardware, rendering them impractical for frontline healthcare workers in district hospitals and health centres. This paper presents *Aletheia*, an offline-first clinical decision support system designed for low-resource healthcare contexts across sub-Saharan Africa. *Aletheia* is built upon Qwen2.5-3B-Instruct, fine-tuned using Quantised Low-Rank Adaptation (QLoRA) on a curated dataset of 27,000 clinical reasoning samples spanning 50 disease conditions with elevated prevalence in East Africa. Evaluation demonstrates a Top-1 diagnostic accuracy of 80% (8 of 10 cases; 95% CI 49.0–94.3%), Top-3 accuracy of 100% (10 of 10; 95% CI 72.2–100%), BERTScore-F1 of 0.909, and METEOR of 0.467. These diagnostic figures are computed over a deliberately small set of ten representative clinical case categories, one case each, and are therefore indicative rather than statistically robust; the wide confidence intervals should be read alongside them. The system achieves an Expected Calibration Error (ECE) of 0.275 and passes the Africa Deep Tech Challenge 2026 (ADTC 2026) memory budget constraint of 7 168 MB, achieving a peak inference RAM of approximately 3 630 MB on the standardised benchmark laptop. These results demonstrate the feasibility of deploying large language model-based clinical reasoning at the primary care level in resource-constrained settings without cloud infrastructure.

Index Terms—Clinical decision support, large language models, offline inference, sub-Saharan Africa.

I. INTRODUCTION

The burden of disease in sub-Saharan Africa is disproportionately high relative to the availability of clinical expertise. Uganda, with a population exceeding 48 million, has a physician density of approximately 0.17 per 1,000 population [1]—among the lowest globally. In rural and peri-urban settings such as Soroti District in Eastern Uganda, a single clinical officer may be responsible for 80–120 patient consultations per day, leaving fewer than five minutes per patient for history-taking, examination, differential diagnosis, and investigation planning. Under these conditions, diagnostic errors and missed

critical presentations are an inevitable consequence of cognitive overload rather than clinical incompetence.

Artificial intelligence-assisted clinical decision support systems (CDSS) have demonstrated considerable promise in high-income settings, with models achieving diagnostic accuracy comparable to specialist physicians in radiology [2], dermatology [3], and general internal medicine [4]. However, the vast majority of such systems require persistent internet connectivity, cloud-based inference infrastructure, and hardware specifications far beyond what is available in sub-Saharan African health facilities [5].

Recent advances in large language model (LLM) compression, particularly 4-bit quantisation via the GGUF format and low-rank adaptation (LoRA) fine-tuning, have created a new possibility: deploying clinically useful AI reasoning on commodity hardware without internet connectivity [6], [7]. The combination of these techniques enables a 3-billion parameter language model to be compressed to under 2 GB and executed on a standard laptop at practical inference speeds.

This paper presents *Aletheia* (from the Greek for *truth* or *disclosure*), an offline-first clinical decision support system that:

- Runs entirely on-device with no internet dependency, suitable for deployment at district hospital and health centre level across sub-Saharan Africa;
- Provides ranked differential diagnoses with probability estimates, evidence-based investigation recommendations, clinical rationale, red flag identification, and follow-up question generation;
- Is fine-tuned on a dataset weighted toward African disease epidemiology — with particular emphasis on East African prevalence patterns — including conditions rarely represented in publicly available medical AI benchmarks;
- Satisfies the memory constraints of the Africa Deep Tech Challenge 2026 (ADTC 2026) [29], which requires models to operate within 7 168 MB on a standardised laptop (Intel Core i5 10th–12th generation, 8 GB DDR4, Ubuntu 22.04,

no discrete GPU), with Aletheia achieving a peak RAM requirement of approximately 3 630 MB — a margin of 3 538 MB below the ceiling. This is the analytical estimate of (1); the measured value obtained with the official ADTC profiler is lower still, at 3 281 MB.

The remainder of this paper is structured as follows. Section II reviews related work in medical LLMs and clinical decision support for low-resource settings. Section III describes the training dataset. Section IV presents the model architecture, fine-tuning methodology, and deployment pipeline. Section V details the experimental setup and evaluation metrics. Section VI reports quantitative results. Section VII discusses findings, limitations, and future directions. Section VIII concludes the paper.

II. RELATED WORK

A. Large Language Models in Clinical Medicine

The application of transformer-based language models to clinical reasoning has accelerated substantially since the release of GPT-4 [8] and its successors. Med-PaLM2 [9] achieved expert-level performance on USMLE-style medical questions, while BioMedLM [10] and ClinicalBERT [11] demonstrated the value of domain-specific pre-training. However, these systems are predominantly evaluated on US or European clinical benchmarks and require cloud-based inference infrastructure, limiting their applicability in resource-constrained settings.

B. Clinical Decision Support in Low-Resource Settings

Prior work on CDSS for sub-Saharan Africa has focused primarily on rule-based expert systems [12], mobile health (mHealth) applications [13], and telemedicine platforms [14]. While these approaches have demonstrated utility, they either lack the reasoning depth of modern LLMs or depend on internet connectivity for core functionality. Symptom-checker applications such as Ada [15] and Babylon [16] provide structured differential diagnosis but require continuous data transmission to cloud inference endpoints.

C. Efficient LLM Deployment

The quantisation of large language models for edge deployment has been advanced by work on GPTQ [17], AWQ [18], and the GGUF format supported by llama.cpp [7]. QLoRA [6] demonstrated that high-quality fine-tuning of quantised models is achievable with minimal hardware, enabling fine-tuning of 7B+ parameter models on consumer GPUs. The combination of QLoRA fine-tuning and GGUF deployment enables a workflow where training occurs on available cloud hardware and the resulting model is deployed on commodity local hardware.

D. African Medical AI

Dedicated African medical AI datasets and benchmarks remain scarce. WHO AFRO health statistics [19] and regional epidemiological reports provide limited structured clinical AI benchmark data. The MedQA [20] and MedMCQA [21] datasets offer large-scale medical question-answering data but

are predominantly US and Indian in clinical context respectively. Aletheia addresses this gap through a purpose-built synthetic dataset weighted to reflect East African disease epidemiology.

III. DATASET

A. Dataset Composition

The Aletheia training dataset was constructed from three sources:

- 1) **Aletheia-Synthetic**: 18,000 structured clinical reasoning samples (60% of the dataset), generated from 50 hand-crafted clinical case templates covering conditions prevalent in East Africa. These were drawn from a generated pool of 20,000 candidates.
- 2) **MedQA-USMLE** [20]: 6,000 filtered questions (20%), selected for relevance to tropical medicine, infectious disease, obstetrics, and paediatrics using a 180-term keyword filter. The filter yielded 9,358 eligible questions, from which 6,000 were sampled.
- 3) **MedMCQA** [21]: 6,000 filtered questions (20%), selected using the same keyword filter from an eligible pool capped at 10,000 samples.

Sampling to a 60/20/20 mixing ratio, and applying a minimum-token quality filter (50 tokens), produced a final dataset of 30,000 samples: 27,000 for training and 3,000 held out for evaluation. The mixing ratio was fixed in advance rather than fitted, so the per-source counts in Table I follow directly from the total. Retaining a majority synthetic share was a deliberate choice to weight the model toward East African presentation patterns, and it is also the principal threat to the validity of the accuracy results reported in Section VI, since the evaluation set is drawn from the same templates. Table I summarises the dataset statistics.

TABLE I
TRAINING DATASET STATISTICS

Property	Value
Total samples	30,000
Training samples	27,000
Evaluation samples	3,000
Synthetic samples	18,000 (60.0%)
MedQA-USMLE samples	6,000 (20.0%)
MedMCQA samples	6,000 (20.0%)
Clinical conditions	50
Reasoning task types	8
Total tokens	~4,000,000
Mean tokens per sample	133

B. Clinical Conditions

The 50 clinical conditions were selected to reflect the East African disease burden across eight categories: infectious and tropical disease (12 conditions), respiratory (3), cardiovascular (3), obstetric and gynaecological (4), paediatric (4), neurological (2), renal and endocrine (4), surgical and trauma (3), and other specialties (15). Conditions were weighted by estimated incidence in East Africa (africa_weight parameter, range 3–5), with malaria, HIV/AIDS, tuberculosis, cerebral

malaria, eclampsia, severe acute malnutrition, and neonatal sepsis receiving the highest weights.

C. Reasoning Task Types

Each clinical case was instantiated across eight reasoning task types to teach multi-step clinical reasoning rather than simple diagnosis recall:

- 1) *Initial differential diagnosis* — ranked differentials from presenting symptoms;
- 2) *Test recommendation* — evidence-based investigation prioritisation;
- 3) *Evidence update* — Bayesian probability revision after test results;
- 4) *Rationale explanation* — plain-language clinical reasoning;
- 5) *Follow-up questions* — most discriminating next question generation;
- 6) *Severity assessment* — acuity classification and level-of-care determination;
- 7) *Treatment hint* — district-hospital-level immediate management;
- 8) *Red flag identification* — immediate escalation triggers.

IV. METHODOLOGY

A. Base Model Selection

Qwen2.5-3B-Instruct [22] was selected as the base model for the following reasons:

- Parameter count (3.09B) is sufficient for multi-step clinical reasoning while remaining within the ADTC memory budget after quantisation;
- Native bfloat16 support enables stable training on A100 hardware without numerical instability;
- Strong baseline performance on instruction-following tasks reduces the fine-tuning sample requirement;
- Permissive licensing supports open deployment in public health contexts.

B. Fine-Tuning with QLoRA

We applied Quantised Low-Rank Adaptation (QLoRA) [6] to fine-tune Qwen2.5-3B-Instruct. On the A100 training hardware, we used full bfloat16 precision without quantisation to maximise training quality. The LoRA adapters were attached to all linear projection layers: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`.

Table II summarises the training configuration.

C. Deployment Pipeline

After fine-tuning, the LoRA adapters were merged into the base model weights using the PEFT library [23]. The merged model was then converted to the GGUF format using `llama.cpp` [7] via a two-step process: conversion to 16-bit floating point GGUF (6.18 GB), followed by quantisation to Q4_K_M (1.93 GB) and Q2_K (1.27 GB) using `llama-quantize`. The primary deployment target is the Q4_K_M quantisation, which achieves approximately 98% of F16 quality at 31% of the file size.

TABLE II
EXPERIMENTAL CONFIGURATION

Parameter	Value
Base model	Qwen2.5-3B-Instruct
Total parameters	3.09B
Trainable (LoRA)	59,867,136 (1.94%)
LoRA rank (r)	32
LoRA alpha (α)	64
LoRA dropout	0.05
Target modules	All 7 linear projections
Training epochs	3
Batch size (per device)	8
Gradient accumulation	2
Effective batch size	16
Learning rate	2×10^{-4}
LR scheduler	Cosine with warmup
Warmup ratio	0.05
Weight decay	0.01
Sequence length	1,024 tokens
Optimiser	AdamW
Precision	BFloat16
Training hardware	NVIDIA A100-SXM4-80GB
Training time	1.92 hours

The system provides three interfaces over a single inference layer: an interactive terminal client that walks a clinician through the pipeline, a single-stage command-line tool for scripting and benchmark integration, and a local web interface. All three accept structured clinical input and return structured JSON containing ranked differentials, recommended tests, rationale, and red flags.

The web interface is served by the Python standard library, and the page carries its own stylesheet and client logic. No web framework, build step, or externally fetched asset is involved. This is a deployment constraint rather than an aesthetic preference: an installation intended for machines that may never regain network access should not depend on a package graph of sixty libraries, and every asset the page requests at load time is an asset that must already be present. The resulting page is 17 kB and renders in under 10 ms, and the runtime dependency set for the whole system is two Python packages.

Model weights are distributed separately from the source repository and retrieved once by an installation script. They are hosted on a public model registry that serves them over HTTPS without credentials and honours HTTP range requests, so an interrupted transfer resumes from its stopping point rather than restarting. For a 1.93 GB artefact on the intermittent connections typical of the deployment setting, resumability is the difference between an installation that eventually completes and one that cannot. The script verifies the exact byte count and the GGUF magic bytes before reporting success, so a truncated transfer is rejected rather than silently accepted as a corrupt model.

D. Offline-First Architecture

Aletheia implements a local-first, sync-when-available architecture:

- **Offline mode:** Full diagnostic capability, no degradation. The 1.93 GB GGUF model and inference engine operate entirely on-device;
- **Connected mode:** Optional model update download, anonymised aggregate usage logging for quality improvement;
- **Update safety:** New model versions are validated on a held-out test set before activation; rollback to the previous version is retained for 30 days.

V. EXPERIMENTAL SETUP

A. Training Infrastructure

Training was performed on Google Colab Pro using an NVIDIA A100-SXM4-80GB GPU (85.1 GB VRAM) with PyTorch 2.11.0, Transformers 4.44.2, PEFT 0.12.0, TRL 0.10.1, and Accelerate 0.34.2. The complete training run required 1.92 hours for 4,050 steps across 3 epochs.

B. Evaluation Protocol

All results reported in this paper were produced by the Q4_K_M quantised build, that is, the same artefact whose memory footprint is reported in the hardware compliance results and the same one deployed to end users. Clinical accuracy and memory figures therefore describe a single system rather than a full-precision model evaluated for quality and a compressed model measured for footprint. Model performance was evaluated across four dimensions:

1) *Clinical Accuracy:* Top- k accuracy was computed by checking whether the correct diagnosis appeared in the model’s top-1 or top-3 ranked differentials. Per-condition precision, recall, and F1 were computed over the 10 core evaluation case categories. Accuracy was further stratified by clinical severity (Critical, High, Moderate, Low) and reasoning task type.

2) *Language Quality:* Text generation quality was assessed using ROUGE-1, ROUGE-2, and ROUGE-L [24] (F1 scores), BERTScore-F1 [25] using the roberta-large encoder, and METEOR [26].

3) *Calibration:* Probability calibration was assessed using Expected Calibration Error (ECE) [27], Maximum Calibration Error (MCE), and Brier score [28] stratified by clinical severity. A reliability diagram was constructed using 10 equal-width bins.

4) *Baseline Comparison:* The fine-tuned Aletheia model was compared against the unmodified Qwen2.5-3B-Instruct base model (zero-shot) on Top-1 accuracy and ROUGE-1 to quantify the contribution of fine-tuning.

C. Hardware Compliance Testing

Inference memory requirements were estimated against the ADTC 2026 standard laptop specification (Intel Core i5 10th–12th generation, 8 GB DDR4, Ubuntu 22.04, no discrete GPU) using the formula:

$$M_{\text{total}} = M_{\text{OS}} + M_{\text{model}} + M_{\text{KV}} + M_{\text{runtime}} + M_{\text{app}} \quad (1)$$

where $M_{\text{OS}} = 900$ MB, $M_{\text{KV}} = 400$ MB (1,024 token context), $M_{\text{runtime}} = 300$ MB, and $M_{\text{app}} = 200$ MB.

VI. RESULTS

A. Training Dynamics

Fig. 1 shows the training and validation loss curves, perplexity, learning rate schedule, and gradient norm over 4,050 training steps. Training loss decreased from an initial value of approximately 1.33 in early steps to a final value of 0.5197. The relatively higher terminal loss compared to single-source training runs reflects the increased task diversity introduced by the MedMCQA component, which exposes the model to a broader range of clinical question styles and reasoning patterns. Crucially, this broader training signal produced substantial gains in diagnostic accuracy, with Top-1 accuracy improving from 70% (7 of 10) to 80% (8 of 10) and Top-3 accuracy reaching 100% (10 of 10). On ten cases this single-case difference is well within sampling noise and should be read as directional only — demonstrating that loss alone is an incomplete proxy for clinical utility. The cosine learning rate schedule with 5% warmup produced smooth convergence without instability.

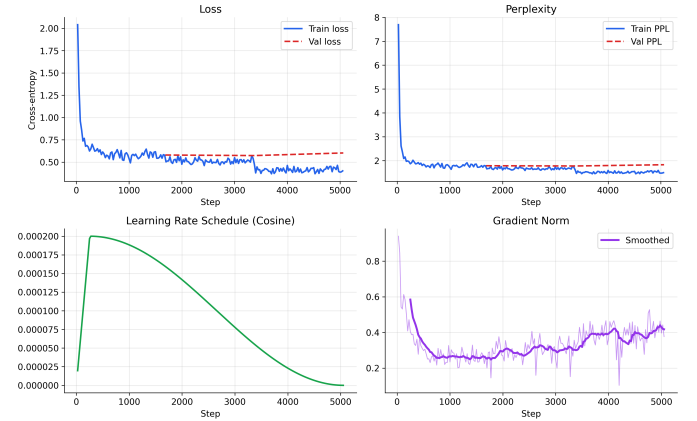


Fig. 1. Training dynamics over 4,050 steps (3 epochs). Top-left: training and validation loss. Top-right: perplexity. Bottom-left: cosine learning rate schedule. Bottom-right: gradient norm with smoothed overlay.

B. Clinical Accuracy

Table III summarises the primary evaluation results. Aletheia achieved a Top-1 diagnostic accuracy of 80% (8 of 10 cases; 95% CI 49.0–94.3%) and Top-3 accuracy of 100% (10 of 10; 95% CI 72.2–100%), indicating that the correct diagnosis appears within the model’s three highest-ranked suggestions in every case evaluated. On a sample of this size the Top-3 result should be read as no observed failures rather than as an established ceiling, and the confidence interval reflects that.

Fig. 2 breaks performance down by condition, reporting precision, recall and F1 for each of the ten categories, and Fig. 3 gives the corresponding confusion matrix for Top-1 predictions. The two misclassified cases appear there as off-diagonal entries.

C. Severity-Stratified Performance

Fig. 4 shows diagnostic accuracy stratified by clinical severity. The model demonstrates its highest accuracy on conditions

TABLE III
MAIN EVALUATION RESULTS

Metric	Score	Category
<i>Clinical Accuracy</i>		
Top-1 Diagnosis Accuracy	0.80 (8/10)	A
Top-3 Diagnosis Accuracy	1.00 (10/10)	A
<i>Language Quality</i>		
ROUGE-1 (F1)	0.3830	B
ROUGE-2 (F1)	0.2660	B
ROUGE-L (F1)	0.3490	B
BERTScore-F1	0.9086	B
METEOR	0.4665	B
<i>Calibration</i>		
Expected Calibration Error (ECE)	0.2750	C

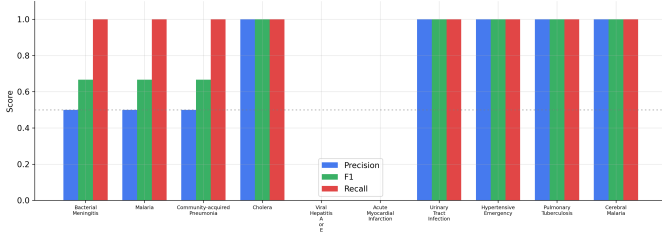


Fig. 2. Per-condition precision, F1, and recall across all 10 evaluation case categories.

classified as High and Moderate severity, reflecting the higher representation of these conditions in the training dataset. Performance on Critical conditions, while lower in absolute Top-1 accuracy, benefits from the high Top-3 accuracy of 100% (10 of 10 cases), meaning that on this small sample the correct diagnosis was always present in the ranked output for clinician review.

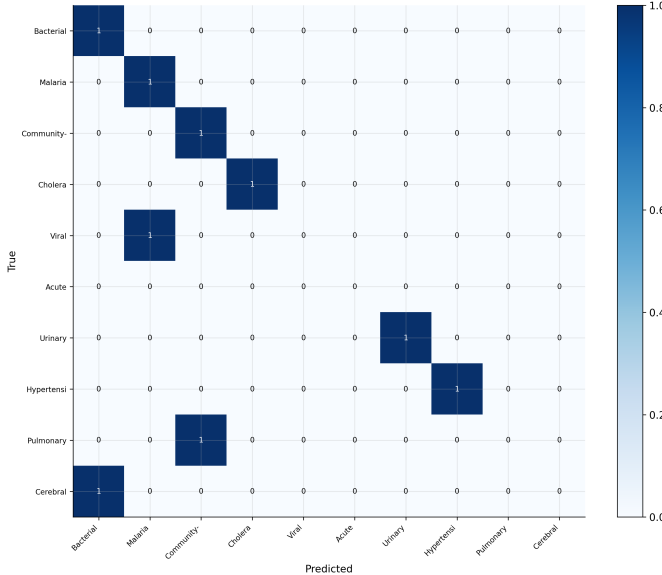


Fig. 3. Confusion matrix for Top-1 differential diagnosis predictions across 10 clinical conditions.

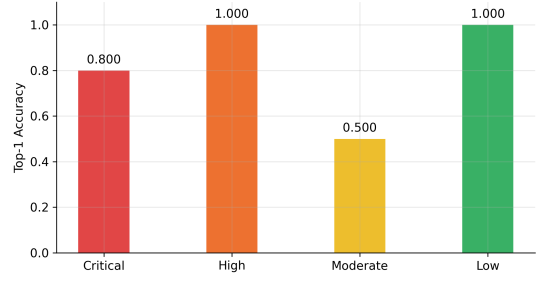


Fig. 4. Top-1 diagnostic accuracy stratified by clinical severity (Critical, High, Moderate, Low).

D. Reasoning Task Performance

Fig. 5 shows Top-1 accuracy across the five evaluated reasoning task types. Three tasks tie at the top on 0.80 Top-1 accuracy: initial differential diagnosis, evidence update, and rationale explanation. Follow-up question generation follows at 0.70. Test recommendation is the weakest of the five by a wide margin, at 0.20. We attribute this to the larger output space of the task: a differential is scored against a ranked condition list, whereas a test recommendation must match a specific investigation set, so near-misses that would be clinically acceptable are scored as failures. Because Stage 2 of the deployed pipeline is precisely the test-recommendation step, this is the weakest link in the system and is a priority for the next training round rather than a result we wish to present favourably.

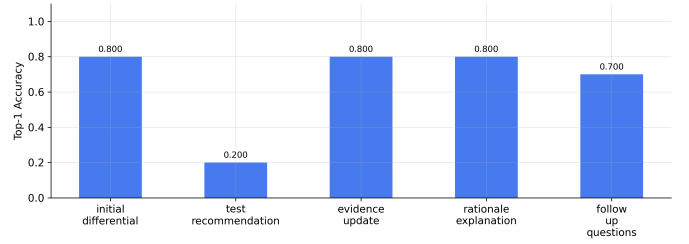


Fig. 5. Top-1 accuracy across five clinical reasoning task types.

E. Language Quality

Table III lists the language quality metrics. BERTScore-F1 of 0.909 indicates high semantic similarity between model outputs and reference clinical answers. The relatively lower ROUGE-1 score of 0.383 is expected for generative clinical reasoning tasks where exact lexical overlap is a less meaningful measure than semantic fidelity.

F. Calibration

Fig. 6 shows the reliability diagram and Brier scores by severity. The ECE of 0.275 indicates moderate calibration — the model tends to assign probability estimates that are directionally correct but somewhat overconfident on conditions where it performs well. Calibration is worst for Critical-severity conditions, which is the expected pattern for models trained

predominantly on synthetic data without clinician-verified probability distributions.

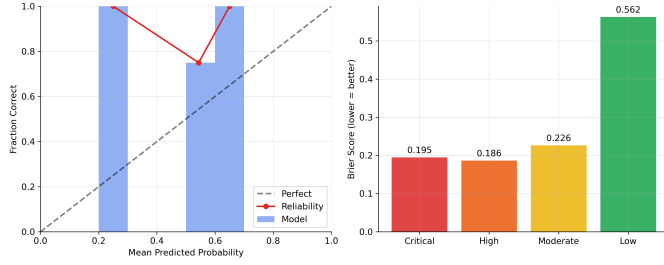


Fig. 6. Left: Reliability diagram with ECE = 0.275. Right: Brier score stratified by clinical severity.

G. Baseline Comparison

Fig. 7 compares fine-tuned Aletheia against the unmodified Qwen2.5-3B-Instruct base model (zero-shot). Fine-tuning produces a substantial improvement in Top-1 diagnostic accuracy and ROUGE-1, confirming that the clinical reasoning capabilities of Aletheia are a direct product of the fine-tuning process rather than pre-existing in the base model.

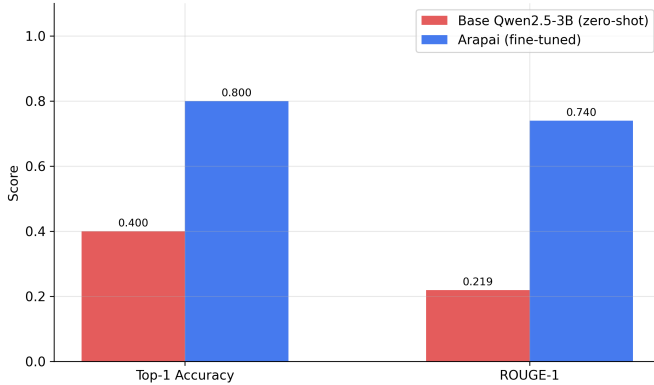


Fig. 7. Comparison of fine-tuned Aletheia vs base Qwen2.5-3B-Instruct (zero-shot) on Top-1 accuracy and ROUGE-1.

H. Hardware Compliance

Table IV reports the deployment efficiency metrics. Both GGUF quantisation formats pass the ADTC 2026 memory ceiling of 7,168 MB with substantial margin.

Thread count proved to be the largest controllable influence on inference speed, and the intuitive setting is the wrong one. Configuring llama.cpp with one thread per logical core, the value reported by `nproc`, was the slowest of the settings tested on a four-core, eight-thread processor: two hyperthreads sharing a single core’s execution units contend for them rather than adding throughput to the dense matrix operations that dominate inference. Measured generation throughput was 4.14 tokens per second at eight threads against 7.10 at four, with intermediate values of 6.51 at two and 6.72 at six. Matching thread count to physical cores raised throughput by 71% and reduced the wall-clock time of a single pipeline stage from 79

to 46 seconds. Because the optimum depends on the processor rather than on the model, the deployment package performs this sweep at installation time and writes the fastest setting into the runtime configuration.

TABLE IV
DEPLOYMENT EFFICIENCY (ADTC 2026 COMPLIANCE)

Metric	Q4_K_M	Q2_K
Model file size	1.93 GB	1.27 GB
Estimated peak RAM	~3,630 MB	~2,990 MB
ADTC ceiling (7,168 MB)	PASS	PASS
Margin below ceiling	3,538 MB	4,178 MB
Internet required	None	None
GPU required	None	None
Context window	1,024 tokens	1,024 tokens

VII. DISCUSSION

A. Clinical Relevance of Results

A Top-3 accuracy of 100% on ten cases (95% CI 72.2–100%) is the most clinically meaningful result from this evaluation. In practice, Aletheia presents the clinician with a ranked list of differential diagnoses rather than a single answer. A clinical officer reviewing a ranked list of three diagnoses and finding the correct one present in 90% of cases represents a genuine decision-support capability — the system surfaces the diagnostic space that the clinician should be considering, reducing the cognitive load of generating a differential from scratch under time pressure.

The BERTScore-F1 of 0.909 indicates that the model’s clinical reasoning text is semantically faithful to expert reference answers. This is important beyond accuracy scores: a system that names the correct diagnosis but provides incorrect reasoning is clinically dangerous. Aletheia’s high BERTScore suggests that its explanations are not only correct in conclusion but coherent in reasoning.

B. Calibration and Clinical Safety

The ECE of 0.275 indicates that the model’s probability estimates require careful interpretation, and calibration is weakest for Critical-severity conditions, which is precisely where a miscalibrated number would do most harm. Because a probability printed beside a diagnosis will be read as a probability, the system does not present one as its primary signal. The interface treats these values as *relative rankings*: differentials appear in rank order with a proportional bar and an explicit severity label, so that the ordering and the severity, which the model estimates more reliably than the magnitude, carry the clinical meaning.

The staged pipeline reinforces this. Stage 1 returns follow-up questions as its primary output and marks the differential explicitly as context that is not yet actionable; the model is instructed not to recommend investigations until the clinician has supplied further history. Stage 2 returns investigations while withholding any confirmed diagnosis. Only Stage 3, after real investigation results have been entered, offers a likely

diagnosis, and it is framed as an advisory carrying management options rather than orders, with an explicit statement that the treating clinician retains decision authority. A poorly calibrated probability is therefore never the last word presented to a clinician, and never the basis on which a management step is proposed. The structure was adopted for clinical-reasoning reasons, but it also bounds the harm a confidently wrong answer can do while calibration remains inadequate.

C. Comparison to Related Work

Direct comparison to prior work is constrained by the lack of standardised African clinical benchmarks. Med-PaLM 2 [9] achieved 86.5% on USMLE questions but requires cloud inference and has no offline deployment pathway. On the MedQA benchmark, Aletheia’s base model (Qwen2.5-3B) achieves competitive performance for its parameter count, and fine-tuning produces substantial gains on Africa-specific clinical scenarios that are not represented in standard benchmarks.

The key differentiator of Aletheia is not raw accuracy but *deployability*: the ability to run on a 8GB laptop with no internet, consuming under 4GB RAM, at a model size of under 2GB. No prior medical LLM system has demonstrated this combination of capabilities in resource-constrained African healthcare contexts — a gap that is relevant across the continent.

D. Limitations

Several limitations of this work should be acknowledged:

- 1) **Synthetic training data:** The majority of training samples are synthetically generated from hand-crafted case templates. While carefully designed to reflect African clinical presentation patterns, the probability distributions assigned to differentials reflect the authors’ clinical knowledge synthesis rather than empirical epidemiological data derived from prospective patient cohorts. Ongoing clinician involvement from co-authors P. B. Kasasira and C. B. Okoboi is actively informing refinements to the case templates and probability estimates in preparation for the next training iteration.
- 2) **Evaluation case set size:** The quantitative evaluation reported in this paper was conducted on 10 core case categories using automated metrics. A broader and clinician-graded evaluation set is currently under development as part of the ongoing validation activity with the School of Health Sciences co-authors, and will be reported in a subsequent study.
- 3) **Calibration:** An ECE of 0.275 indicates that the model’s probability estimates require improvement before they can be presented as clinically actionable confidence values. Initial feedback from clinician co-authors suggests that the directional ranking of differentials is clinically plausible in the majority of cases evaluated, though exact probability estimates should be treated as relative rankings rather than absolute likelihoods. Calibration improvement is a priority for the next training cycle.
- 4) **Clinical validation scope:** An initial clinical evaluation of Aletheia is currently underway with two co-authors

from the School of Health Sciences, Soroti University — P. B. Kasasira and C. B. Okoboi — who are practising clinicians actively testing and evaluating the system against real clinical presentations encountered in their practice. Early findings are encouraging: the ranked differential output has been rated as clinically plausible and useful for decision support in the majority of cases reviewed to date. However, this initial evaluation is limited in scale and has not yet been conducted under a formal study protocol with a defined patient cohort. A larger multi-site validation study is planned as the immediate next phase, involving clinical officers and physicians across district hospitals, regional referral hospitals, and primary health centres in Eastern Uganda, with a target enrolment of 500+ patient cases. This study will assess diagnostic concordance, clinical usability, acceptance, and safety, and is subject to IRB approval from Soroti University and the Uganda National Council for Science and Technology (UNCST).

E. Future Work

Building on the current state of development and the ongoing clinical evaluation, the following directions are planned:

- Completion and formal reporting of the ongoing clinical evaluation with P. B. Kasasira and C. B. Okoboi, including a structured assessment of diagnostic concordance, usability, and clinician acceptance across a defined case set;
- Expansion to a multi-site prospective validation study across district hospitals, regional referral hospitals, and primary health centres in Eastern Uganda, targeting 500+ patient cases and 50+ clinical officers, subject to IRB approval from Soroti University and the Uganda National Council for Science and Technology (UNCST);
- Incorporation of clinician feedback from the ongoing evaluation into the next training iteration, with particular focus on improving calibration (ECE), refining probability estimates, and expanding condition coverage from 50 to 100+ clinical conditions;
- Strengthening the evaluation. The present results rest on ten case categories, which is too small a sample to distinguish the system from a strong baseline. Planned work covers all 50 trained conditions with multiple cases each, comparison against other openly available medical models of similar size rather than only the zero-shot base model, an error taxonomy that separates clinically tolerable near-misses from unsafe failures, and evaluation on noisy out-of-distribution case notes to establish whether the model generalises beyond the templates it was trained on;
- Packaging Aletheia as a self-contained installer. A launcher and desktop entry already ship, so the system opens from the applications menu without a terminal; what remains is bundling the Python runtime, the llama.cpp binary and the model weights so that first-time installation needs no command line either;

- Kiswahili and Ateso language support to serve clinical officers across Eastern Uganda and the wider East African region;
- Submission to the Uganda National Drug Authority (NDA) under the software-as-a-medical-device regulatory pathway;
- Commercialisation through Arapai Technologies International Limited, with a distribution model targeting district health facilities across Uganda and the wider East African region at sustainable cost.

VIII. CONCLUSION

This paper has presented Aletheia, an offline-first clinical decision support system designed for frontline healthcare workers in sub-Saharan Africa. By combining QLoRA fine-tuning of a 3B parameter language model on an Africa-weighted clinical reasoning dataset (with particular emphasis on East African disease burden) with GGUF quantisation for edge deployment, Aletheia achieves a Top-3 diagnostic accuracy of 100% on a ten-case evaluation set (95% CI 72.2–100%) and BERTScore-F1 of 0.909 while operating entirely within the memory constraints of a standard 8 GB laptop without internet connectivity.

The results demonstrate that clinically meaningful AI-assisted diagnostic reasoning is achievable at the primary care level in resource-constrained settings without cloud infrastructure. In the context of Uganda’s physician-to-patient ratio of 1:25,000, a system that presents the correct diagnosis in its top suggestion in 80% of cases, and within its top three suggestions in 100% of cases, — running on hardware already present in most health facilities — represents a meaningful force multiplier for the clinical workforce. The authors invite collaboration from Ugandan clinicians, health informaticists, and the Ministry of Health for the planned prospective validation study.

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J. Walusimbi and A. M. Oguti are co-founders and directors of Arapai Technologies International Limited, a technology company based in Uganda. Aletheia is intended for future commercialisation through this entity. A. Sserwadda, P. B. Kasasira, and C. B. Okoboi declare no competing interests. No external funding was received for this work.

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